ALICE L WALTON SCHOOL OF MEDICINE

Dilated Cardiomyopathy

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Adapted from: Dilated Cardiomyopathy

Listen to this Brick
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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Define dilated cardiomyopathy (DCM).
2. Differentiate the clinical presentation of DCM.
3. Explain the pathophysiology of DCM.
4. Interpret clinical findings to properly diagnose DCM.
5. Outline the

CASE CONNECTION

YT is a 50-year-old female who has been healthy her whole life. About 1 week ago she had a “virus” with fever and fatigue, and about 2 days ago she developed worsening shortness of breath and swelling in her feet and ankles. She now comes to the clinic, wondering if this might be COVID. On exam, severe edema is noted. On auscultation, crackles in the chest are heard and an S_3 heart sound is noted. The chest x-ray is very concerning, because the heart appears to take up about three quarters of the pleural cavity.

What is the likely diagnosis, and what would you tell YT? Consider your answer as you read, and we’ll revisit YT at the end of the brick.

GO TO CONCLUSION

What Is Dilated Cardiomyopathy? (LO #1)

Cardiomyopathies are diseases that weaken the cardiac muscle, often leading to symptoms of heart failure. They encompass a wide range of disparate conditions, and the overlaps can be confusing. The definition of a cardiomyopathy, as per the European Society of Cardiology in 2008 is, “a myocardial disorder in which the heart muscle is structurally and functionally abnormal in the absence of coronary artery disease, hypertension, valvular disease, and congenital heart disease sufficient to explain the observed myocardial abnormality.”

So, while hypertension, ischemic heart disease (coronary artery disease), and chronic valvular disease can sometimes lead to a dilated heart and heart failure, they are not considered cardiomyopathies because they are not primarily diseases of heart muscle. Just realize that you will still sometimes hear the term “ischemic cardiomyopathy” used when heart failure follows a myocardial infarction, even though it does not meet the definition above.

Cardiomyopathies can be grouped into four types:

- **Dilated cardiomyopathy**
- **Hypertrophic cardiomyopathy**
- **Restrictive cardiomyopathy**
- **Arrhythmogenic right ventricular cardiomyopathy**

Dilated cardiomyopathy (DCM) is the most common type, comprising 90% of all cardiomyopathies, and it is the subject of this brick.

DCM is defined as a cardiomyopathy leading to **dilation** (i.e., stretching and thinning out) of the cardiac chambers (Figure 1). DCM can appear at any age, although it usually appears in people ages 20 to 60 years. In the United States, this condition is seen in as many as three times more males than females, and three times more in Black patients compared with White.

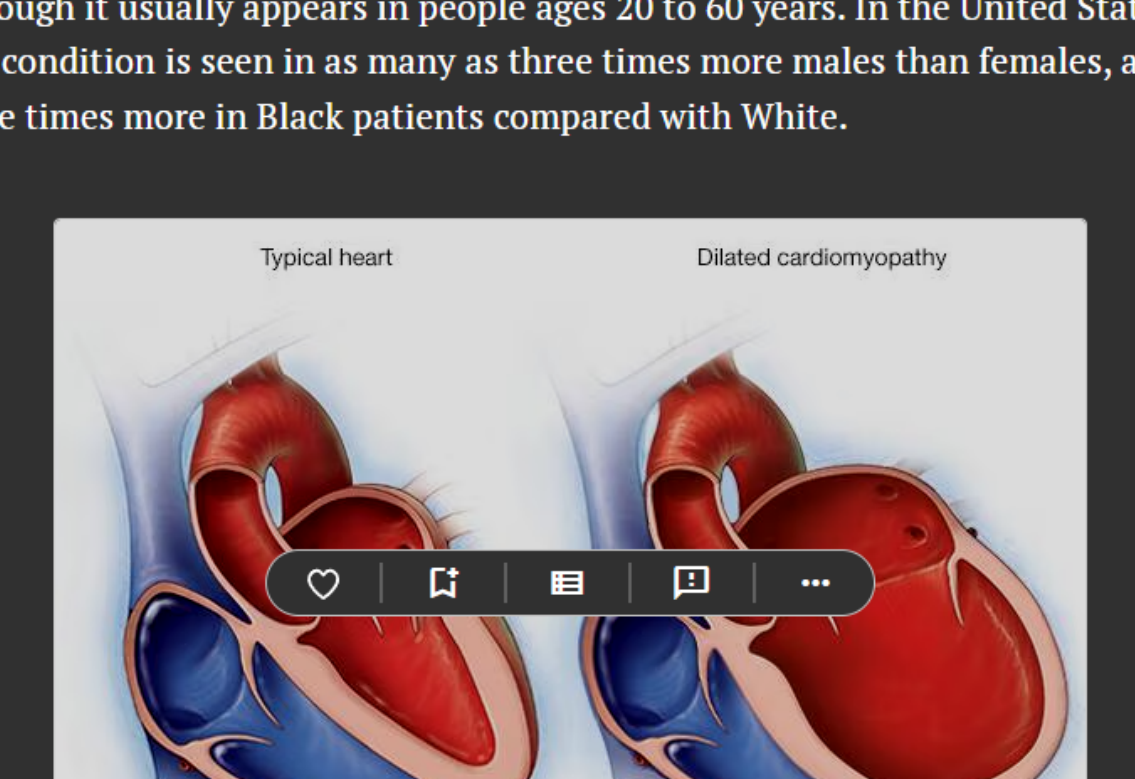

Figure 1. Dilated Cardiomyopathy. DCM is characterized by enlargement of the cardiac chambers and thinning of the ventricular walls.

What is the most common type of cardiomyopathy?

How Do Patients With Dilated Cardiomyopathy Present? (LO #2)

Patients with DCM p left- and right-sided systolic heart failure. Because the left ventricle cannot pump blood forward as effectively, there may be fatigue and dizziness, and pulmonary congestion can occur with concordant signs like orthopnea and paroxysmal nocturnal dyspnea. When the right ventricle becomes involved, the patient may experience lower extremity edema.

Which side of the heart fails commonly in patients with dilated cardiomyopathy?

What Is the Pathophysiology of Dilated Cardiomyopathy? (LO #3)

The dilation of the cardiac chambers in DCM often starts with the left side of the heart, then the dilation spreads to the right ventricle and atria (Figure 2).

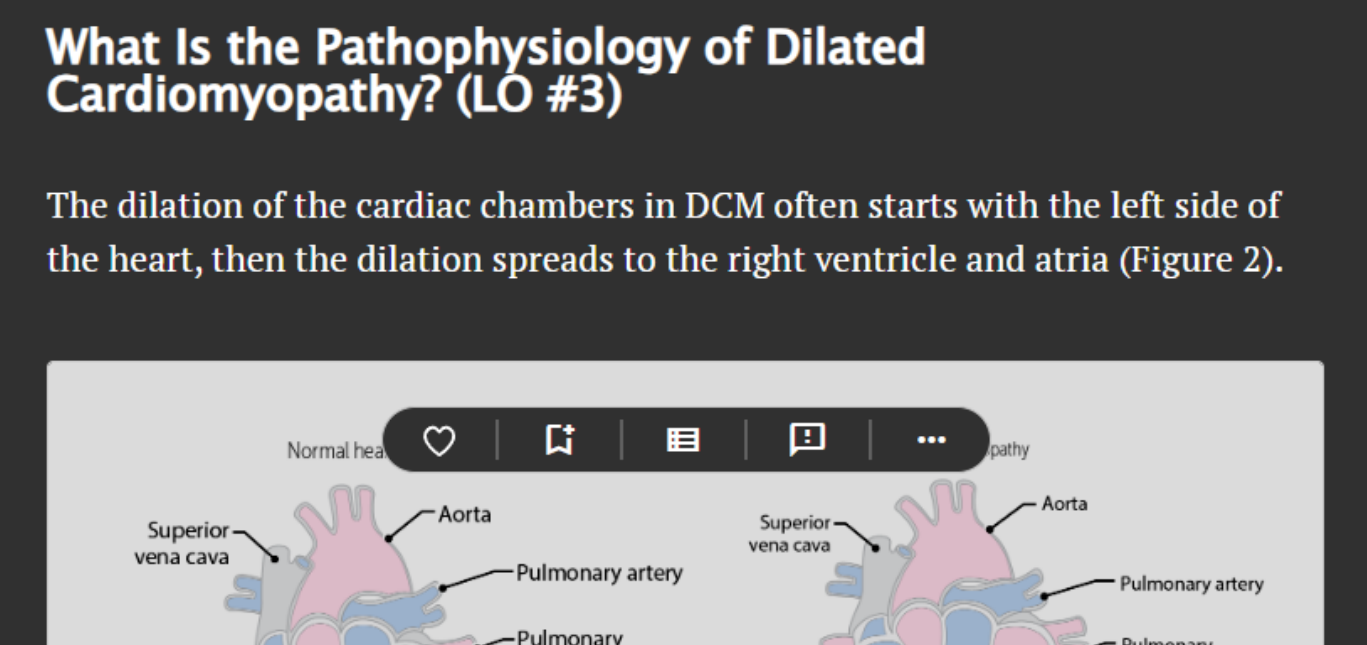

Figure 2. Dilated Cardiac Chambers. In comparison to normal, a patient with DCM will have enlarged (dilated) cardiac chambers, both atria and ventricles, involving both sides of the heart.

Many different causes of heart disease can produce dilation, to include ischemic and valvular disease. But to classify as DCM, the dilation needs to result from a pathologic process involving the heart muscle itself (and not one secondary to another disease). The muscle disorders leading to DCM can be the result of genetic causes, have associations with inflammatory or infectious conditions (e.g., myocarditis), or be the result of toxic damage to the muscle (e.g., drugs and especially **alcohol**). Other cases are idiopathic (i.e., without an identifiable cause). Patients with hypertension, obesity, and thyroid dysfunction may also develop dilated cardiomyopathy.

Risk Factors

A family history of DCM is a risk factor for developing the condition. Most of the rare inherited for however, other inheritance patterns have been observed.

CLINICAL CORRELATION

There are more than 30 mutations associated with DCM. These mutations often occur in sarcomere proteins (basic contractile units of muscle fibers) (ex. *TTN* gene that makes titin protein in sarcomere). Resultant sarcomere dysfunction is the cause of the systolic dysfunction characteristic of DCM.

When DCM is genetically transmitted, what type of inheritance pattern is most commonly observed?

Causes

Idiopathic. By defini se is unknown. Some theories postulate it may be the result of a previous undetected viral infection causing immune-mediated damage to the myocardium. Some of these “idiopathic” cases likely fall into the genetic category, described above, but a genetic diagnosis is never made.

Alcohol. Heavy alcohol drinking, typically for at least 5 years, may lead to DCM. Alcohol appears to be directly toxic to myocardial cells. About one third of patients with heavy alcohol drinking also have liver disease, which can worsen their symptoms of volume overload. Alcohol toxicity of the heart can sometimes reverse with alcohol cessation.

Drugs. Certain medications can cause DCM, such as the chemotherapy drug **doxorubicin**, although the mechanism is not well understood. Other drugs that commonly can lead to DCM are dobutamine, diuretics, ampicillin, tetracycline, and tricyclic antidepressants. Chronic cocaine use may also cause DCM.

Myocarditis. Myocardial inflammation, mostly caused by infection (viral or bacterial) or immune reactions, commonly leads to DCM.

Viral infections are the most common cause of myocarditis, and more than 20 viruses are known to cause myocarditis. The most common viral agents are Coxsackie B, adenovirus, parvovirus, and human herpesvirus 6. Note that the recent COVID-19 pandemic has seen young patients develop myocarditis because of their viral syndrome. We may also find that patients with advanced HIV infection may develop DCM, but the virus is not thought to be directly injurious to myocytes.

How do viruses cause myocardial damage? They may hijack the cellular organelles to reproduce more virus, ultimately killing the cardiomyocytes. Also, the immune system recruits white blood cells to kill its own cardiomyocytes, an effect that can contir

Other infections that may cause myocarditis and DCM include Lyme disease (bacterial; *Borrelia burgdorferi*) and Chagas disease, caused by the parasite *Trypanosoma cruzi*.

CLINICAL CORRELATION

Chagas disease is transmitted by infected triatomine insects, commonly known as "kissing bugs." These nocturnal blood feeders typically bite near the mouth, which gives rise to their nickname. During or shortly after feeding, the insect defecates, and the parasite - concentrated in the feces - can enter the host through the bite wound or adjacent mucosal surfaces.

Chagas disease is the leading cause of death from parasitic infection in the Western Hemisphere, primarily due to its high burden of chronic cardiac disease and associated complications. The disease burden is greatest in rural regions of Latin America.

Unilateral periorbital swelling (Romaña sign) is characteristic of the acute stage of the infection.

Rarely, myocarditis can develop in the absence of infection, as in autoimmune disease. Causes include giant cell myocarditis resulting from T-lymphocyte inflammation and eosinophilic myocarditis caused by hypersensitivity reactions to medications like dopamine or diuretics. In these cases, patients may have rash, fever, and a high blood eosinophil count, alongside symptoms of heart failure.

Peripartum Cardiomyopathy. This is a rare disorder seen in late pregnancy or up to 5 months after delivery. The cause is unknown but may involve cytokines, abnormal angiogenesis, and prolactin.

Iron Overload. This can result from hereditary hemochromatosis or from repeated blood transfusions. It is thought to be a result of disrupting enzymatic systems, or an injury from iron-mediated production of reactive oxygen species.

Thiamine Deficiency. This is also known as wet beriberi when associated with DCM. The deficiency can cause cardiac damage and lead to DCM. It’s uncommon in developed countries, except in malnourished people.

In myocarditis, how do viruses cause myocardial damage?

Gross Pathology

With DCM, the heart ventricular walls are themselves of normal thickness or characteristically thinner than normal. Thus, the increased size of the heart is due to enlargement of the cardiac chambers ("dilation"). In cases of dilated cardiomyopathy, the width of the heart may take up more than 50% of the total chest cavity width. Classically, the heart is described as **spherical** (Figure 3).

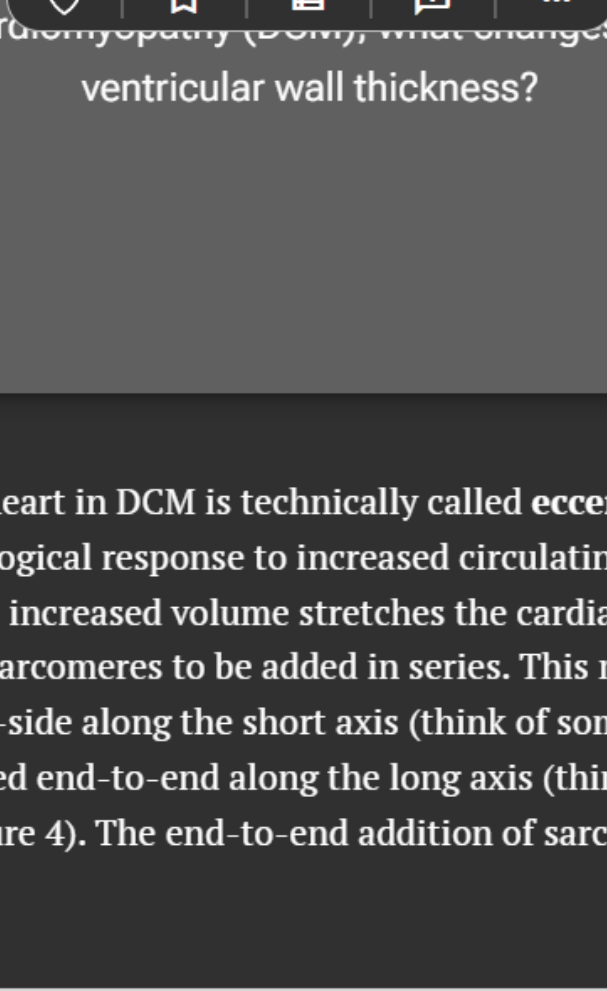

Figure 3. DCM, Cross-Section. Note the ventricular dilation associated with thin ventricular walls. Note also the overall spherical shape of the heart.

In dilated card s occur to the ventricular wall thickness?

The dilation of the heart in DCM is technically called **eccentric hypertrophy**. It’s mainly a physiological response to increased circulating volume in the heart chambers. The increased volume stretches the cardiac muscle cells, which prompts the sarcomeres to be added in series. This means that instead of being added side-by-side along the short axis (think of something getting fatter), they are added end-to-end along the long axis (think of something getting longer) (Figure 4). The end-to-end addition of sarcomeres produces dilation.

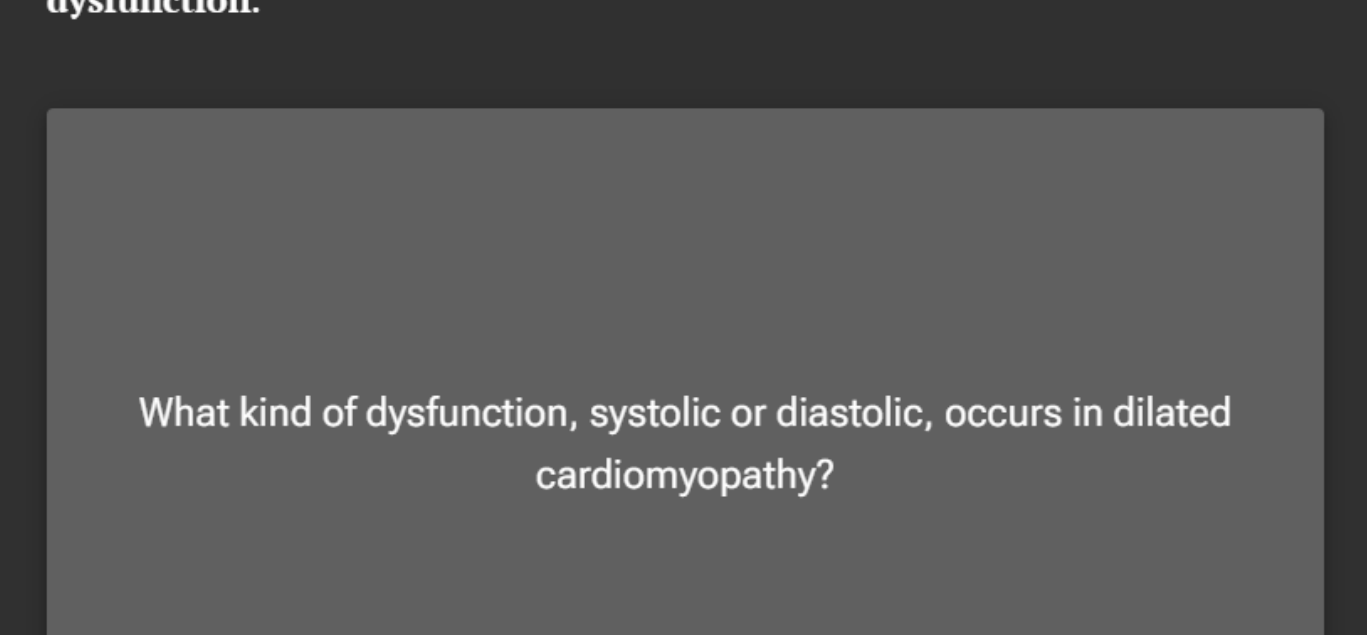

Figure 4. Myocardial Eccentric Hypertrophy.

Unfortunately, the dilated heart, with its in-series sarcomeres, does not pump very well. Adjacent sarcomeres arranged in parallel are better able to generate force, while lined-up sarcomeres in series do not actually improve contraction. All in all, this prevents the heart from contracting properly. This inability to contract properly because of a weak myocardium is termed **systolic dysfunction**.

What kind of dysfunction, systolic or diastolic, occurs in dilated cardiomyopathy?

Histology

Histological findings of the myocardium in noninflammatory causes of DCM are often nonspecific toplasmic vacuolizations, and interstitial fibrosis.

In cases of myocarditis (e.g., viral and bacterial infection), histology will show myocardial inflammatory infiltrates with findings of necrosis. Most cases of viral myocarditis will present with lymphocytic infiltrates (Figure 5). In contrast, bacterial myocarditis will have mostly neutrophilic infiltrates, and myocarditis as part of hypersensitivity reactions will have predominant eosinophilic infiltrates.

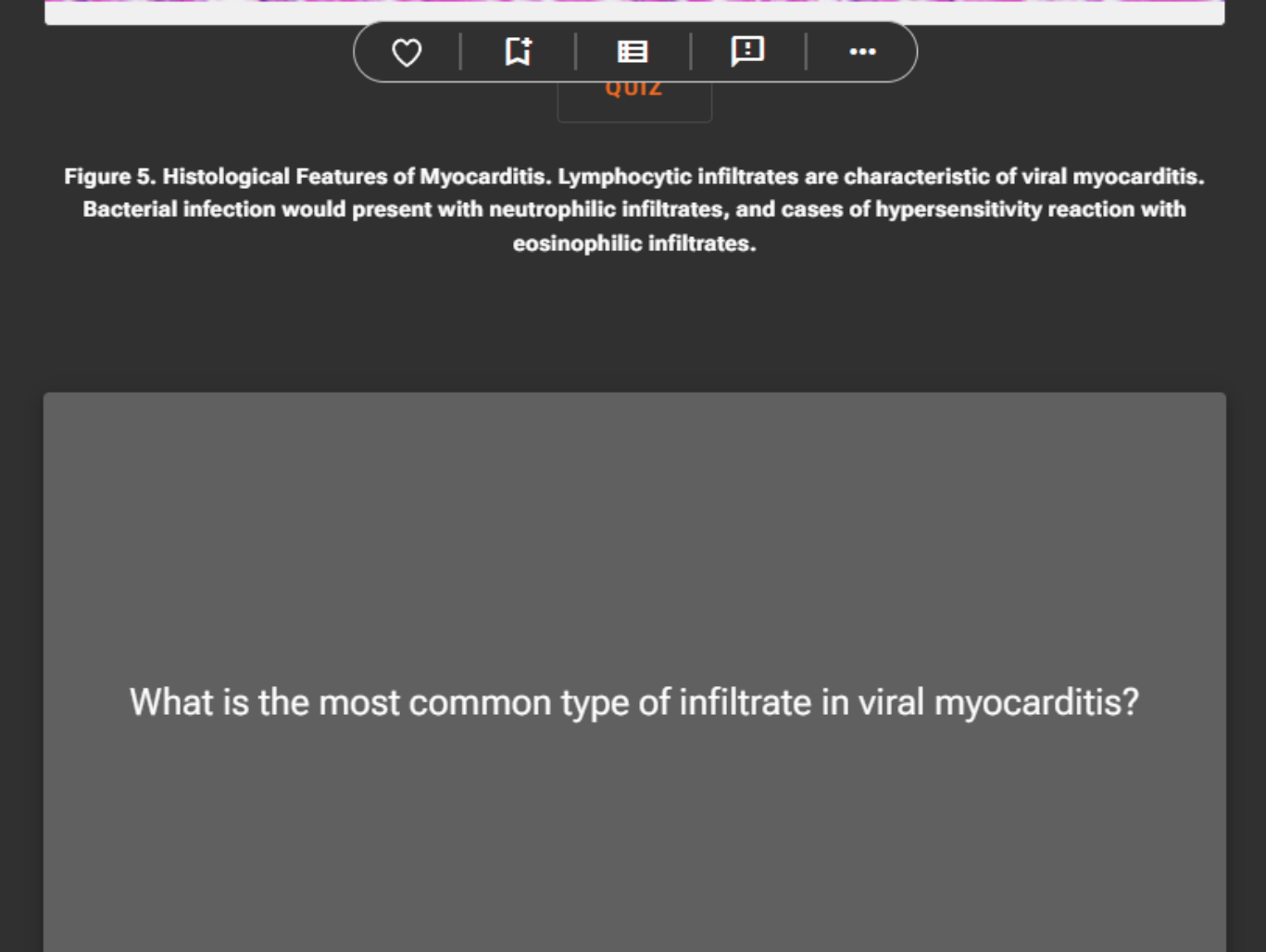

Figure 5. Histological Features of Myocarditis. Lymphocytic infiltrates are characteristic of viral myocarditis. Bacterial infection would present with neutrophilic infiltrates, and cases of hypersensitivity reaction with eosinophilic infiltrates.

What is the most common type of infiltrate in viral myocarditis?

How Do We Diagnose Dilated Cardiomyopathy? (LO #4)

Clinical examination and echocardiography studies are the methods most commonly used to diagnose dilated cardiomyopathy.

Physical Examination

Patients will have typ including pulmonary crackles, an S_3 heart distension (JVD).

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- Subac T. Dilated cardiomyopathy. *MSD Manual. Professional Version*. Updated September 2022. Accessed September 20, 2022. https://www.msdmanuals.com/professional/cardiovascular-disorders/cardiomyopathies/dilated-cardiomyopathy?query=dilated%20cardiomyopathy

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Define dilated cardiomyopathy (DCM).~~
02 ☑ ~~Differentiate the clinical presentation of DCM.~~
03 ☑ ~~Explain the pathophysiology of DCM.~~
04 ☑ ~~Interpret clinical findings to properly diagnose DCM.~~
05 ☑ ~~Outline the management plan of DCM.~~

Objective Confidence: 5/5